

Complete Elliptic Reduction and Two-Phase Return for a Hamiltonian Resonant Triad

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Abstract

For the canonical complex resonant triad we prove global Liouville integrability and reduce every intensity orbit to a real cubic. Every regular orbit is written in Jacobi form with its exact intensity period.

1 Canonical convention and first integrals

On $\mathbb{C}^3$ use

$$\{f, g\} = -i \sum_{j=1}^3 (f_{z_j} g_{\bar{z}_j} - f_{\bar{z}_j} g_{z_j})$$

and the real Hamiltonian

$$H = z_1 z_2 \bar{z}_3 + \bar{z}_1 \bar{z}_2 z_3. \quad (1)$$

Hamilton's equations are

$$i\dot{z}_1 = \bar{z}_2 z_3, \quad i\dot{z}_2 = \bar{z}_1 z_3, \quad i\dot{z}_3 = z_1 z_2. \quad (2)$$

Writing $I_j = |z_j|^2$ and $w = z_1 z_2 \bar{z}_3$ gives

$$\dot{I}_1 = \dot{I}_2 = -2 \operatorname{Im} w, \quad \dot{I}_3 = 2 \operatorname{Im} w.$$

Consequently

$$N_1 = I_1 + I_3, \quad N_2 = I_2 + I_3, \quad H \quad (3)$$

are conserved. Direct evaluation of the displayed bracket gives pairwise involution. On the open set $I_1 I_2 I_3 \sin \Psi \neq 0$, where $\Psi = \arg z_1 + \arg z_2 - \arg z_3$, the first two differentials are independent action differentials and

$$\partial_\Psi H = -2\sqrt{I_1 I_2 I_3} \sin \Psi \neq 0.$$

Thus the three integrals are generically independent. Their common levels are bounded because $0 \leq I_3 \leq \min(N_1, N_2)$ and $I_j = N_j - I_3$ for $j = 1, 2$. Polynomial local existence and boundedness give a global flow on all of $\mathbb{R}$.

2 The real cubic and every regular intensity

Put $x = I_3$. Since $\dot{x} = 2 \operatorname{Im} w$ and $|w|^2 = x(N_1 - x)(N_2 - x)$,

$$\dot{x}^2 = 4x(N_1 - x)(N_2 - x) - H^2 =: P(x). \quad (4)$$

Let $N_- := \min(N_1, N_2)$ and $N_+ := \max(N_1, N_2)$. The function $f(x) = x(N_1 - x)(N_2 - x)$ has one maximum on $[0, N_-]$, at

$$x_* = \frac{N_1 + N_2 - \sqrt{N_1^2 - N_1 N_2 + N_2^2}}{3}, \quad H_{\max} = 2\sqrt{f(x_*)}. \quad (5)$$

For $0 < |H| < H_{\max}$, signs at 0, the maximum, N_- , N_+ , and infinity prove that P has exactly three roots

$$0 < r_1 < r_2 < N_- \leq N_+ < r_3. \quad (6)$$

They obey

$$\sum r_j = N_1 + N_2, \quad \sum_{j < k} r_j r_k = N_1 N_2, \quad r_1 r_2 r_3 = \frac{H^2}{4}. \quad (7)$$

Theorem 1 (regular elliptic orbit). *Let $0 < |H| < H_{\max}$ and define*

$$m = \frac{r_2 - r_1}{r_3 - r_1}, \quad u = \sqrt{r_3 - r_1}(t - t_0).$$

Every intensity orbit on this level is

$$x(t) = r_1 + (r_2 - r_1) \operatorname{sn}^2(u \mid m), \quad T_x = \frac{2K(m)}{\sqrt{r_3 - r_1}}. \quad (8)$$

Proof. On $[r_1, r_2]$ the cubic is $4(x - r_1)(r_2 - x)(r_3 - x)$. Substitute (8) and use $(\operatorname{sn}')^2 = (1 - \operatorname{sn}^2)(1 - m \operatorname{sn}^2)$. The resulting identity is exact. Time translation selects either initial sign of $\dot{x}$, and sn^2 has least real period $2K(m)$. $\square$

The analytic root argument and substitution are the *cubic elliptic owner* of revision round zero.

Source boundary

This paper is a source-local reconstruction, not a priority claim. Manley and Rowe own their energy relations; Armstrong et al. own the coupled optical amplitude setting and explicit three-wave solutions; Kaup, Reiman, and Bers give an authoritative primary treatment of broader resonant three-wave dynamics. The exact software receipts are local implementation checks and are not attributed to those sources.

References

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